

STATISTICS PRACTICALS — SHORT NOTES

Steps | Mathematical Formulas | Excel Functions | Charts

1. Histogram, Average & Variance

► Steps

1. Collect data values $x_1, x_2, \dots, x_n$
2. Compute mean $\bar{x} = \Sigma x_i / n$
3. Find each deviation $(x_i - \bar{x})$
4. Square each deviation $(x_i - \bar{x})^2$
5. Sum all squared deviations
6. Divide by n (pop.) or $n-1$ (sample)
7. Group data into class intervals
8. Count frequency per bin → Histogram

∫ Mathematical Formulas

$$\bar{x} = (1/n) \cdot \Sigma x_i$$
$$\sigma^2 = \Sigma (x_i - \bar{x})^2 / n \quad \text{[Population]}$$
$$s^2 = \Sigma (x_i - \bar{x})^2 / (n-1) \quad \text{[Sample]}$$
$$\sigma = \sqrt{\sigma^2} \quad s = \sqrt{s^2}$$

📊 Excel Functions

=AVERAGE(range) → mean $\bar{x}$
=VAR.P(range) → population σ^2
=VAR.S(range) → sample s^2
=STDEV.S(range) → sample s

📊 Column / Bar Histogram (Bins vs Frequency)

2. Stem and Leaf Plot

► Steps

1. Sort all data in ascending order
2. Split each value into stem & leaf
stem = tens digit (floor($x/10$))
leaf = units digit ($x \bmod 10$)
3. List unique stems vertically
4. Write leaves beside each stem
5. Read shape of distribution
6. Find Min, Max, Median, Range

∫ Mathematical Formulas

Stem = $\lfloor x / 10 \rfloor$ (integer part)
Leaf = $x \bmod 10$
Range = Max - Min
Median = middle value (sorted list)

📊 Excel Functions

=INT(A1/10) → stem
=MOD(A1,10) → leaf
=MAX(range) → maximum
=MIN(range) → minimum

📊 No chart — the display itself is visual

3. Box Plot and Dot Plot

► Steps

1. Sort each dataset ascending
2. Find five-number summary:
Min, Q1, Median, Q3, Max
3. Compute IQR = $Q3 - Q1$
4. Lower fence = $Q1 - 1.5 \times \text{IQR}$
5. Upper fence = $Q3 + 1.5 \times \text{IQR}$
6. Flag any outliers beyond fences
7. Dot plot: mark each value on axis

∫ Mathematical Formulas

Q1 = 25th percentile of data
Q2 = Median (50th percentile)
Q3 = 75th percentile of data
IQR = $Q3 - Q1$
Fences: $Q1 - 1.5 \cdot \text{IQR}$ to $Q3 + 1.5 \cdot \text{IQR}$

📊 Excel Functions

=QUARTILE(range,1) → Q1
=MEDIAN(range) → Q2
=QUARTILE(range,3) → Q3
=QUARTILE(range,3)-QUARTILE(range,1) → IQR

📊 Grouped bar (5-num) + Scatter Dot Plot

4. Binomial Distribution Fitting

► Steps

1. Compute $\bar{x} = \Sigma(x \cdot f) / N$ (mean)
2. Set n = number of trials (given)
3. Estimate $p = \bar{x}/n$, $q = 1 - p$
4. Apply Binomial PMF for $x=0,1\dots n$
5. Expected = $N \cdot P(X=x)$
6. Round expected to nearest integer
7. Apply χ^2 goodness-of-fit test

∫ Mathematical Formulas

$$P(X=x) = C(n,x) \cdot p^x \cdot (1-p)^{n-x}$$
$$C(n,x) = n! / [x! \cdot (n-x)!]$$
$$\text{Mean} = n \cdot p$$
$$\text{Variance} = n \cdot p \cdot q \quad (q = 1-p)$$
$$\chi^2 = \Sigma (O - E)^2 / E$$

📊 Excel Functions

=FACT(n)/(FACT(x)*FACT(n-x)) → $C(n,x)$
=BINOM.DIST(x,n,p,FALSE) → $P(X=x)$
=CHITEST(obs_range, exp_range) → p-value

📊 Clustered bar: Observed vs Expected freq

5. Poisson Distribution Fitting

► Steps

1. Compute $\lambda = \Sigma(x \cdot f) / N$ (mean)
2. Use Poisson PMF for each x value
3. Expected = $N \cdot P(X=x)$
4. Round expected to nearest integer
5. Verify $\Sigma \text{Expected} = N$
6. Apply χ^2 goodness-of-fit test

∫ Mathematical Formulas

$$P(X=x) = e^{-\lambda} \cdot \lambda^x / x!$$
$$e \approx 2.71828 \quad \text{(Euler's number)}$$
$$\text{Mean} = \lambda$$
$$\text{Variance} = \lambda \quad \text{(equal to mean)}$$
$$\chi^2 = \Sigma (O - E)^2 / E$$

📊 Excel Functions

=POISSON.DIST(x, λ , FALSE) → $P(X=x)$
=EXP(- λ)* λ^x /FACT(x) → manual
=CHITEST(obs_range, exp_range) → χ^2 test

📊 Clustered bar: Observed vs Expected freq

6. Regression Analysis (Y on X)

► Steps

1. Compute $\Sigma X, \Sigma Y, \Sigma XY, \Sigma X^2, n$
2. Find slope b using normal equations
3. Find intercept a
4. Write equation: $\hat{Y} = a + bX$
5. Calculate fitted $\hat{Y}$ for each X
6. Residual = $Y - \hat{Y}$ for each point
7. Compute R and R^2 (fit quality)

∫ Mathematical Formulas

$$b = (n \Sigma XY - \Sigma X \cdot \Sigma Y) / (n \Sigma X^2 - (\Sigma X)^2)$$
$$a = \bar{Y} - b \cdot \bar{X} = (\Sigma Y - b \cdot \Sigma X) / n$$
$$\hat{Y} = a + bX$$
$$\text{SSE} = \Sigma (Y - \hat{Y})^2$$
$$R^2 = 1 - \text{SSE}/\text{SST}$$

📊 Excel Functions

=SLOPE(Y_range, X_range) → b
=INTERCEPT(Y_range, X_range) → a
=CORREL(X_range, Y_range) → r
=FORECAST(x, Y_range, X_range) → $\hat{Y}$

📊 Scatter plot + Red regression line ($\hat{Y}$)

7. Sample Correlation Coefficient — Pearson's r

► Steps

1. List all (X_i, Y_i) data pairs
2. Compute $\bar{X} = \Sigma X_i / n$ and $\bar{Y} = \Sigma Y_i / n$
3. Find $\sigma_x = \text{STDEV}(X)$, $\sigma_y = \text{STDEV}(Y)$
4. Standardise: $Z_i(X) = (X_i - \bar{X}) / \sigma_x$
5. Standardise: $Z_i(Y) = (Y_i - \bar{Y}) / \sigma_y$
6. Multiply: $Z_i(X) \cdot Z_i(Y)$ for each pair
7. $r = \Sigma [Z_i(X) \cdot Z_i(Y)] / (n-1)$

∫ Mathematical Formulas

$$r = \Sigma (X_i - \bar{X})(Y_i - \bar{Y}) / [(n-1) \cdot \sigma_x \cdot \sigma_y]$$
$$r = \Sigma Z_i(X) \cdot Z_i(Y) / (n-1)$$
$$r = [n \Sigma XY - \Sigma X \cdot \Sigma Y] / \sqrt{[(n \Sigma X^2 - (\Sigma X)^2)(n \Sigma Y^2 - (\Sigma Y)^2)]}$$

Range: $-1 \leq r \leq 1$

📊 Excel + Interpretation

=CORREL(X_range, Y_range) → r
=PEARSON(X_range, Y_range) → r
📊 Scatter Plot: X_i vs Y_i

Interpretation of $|r|$:

$|r| = 1.0$ Perfect correlation
 $0.7 \leq |r| < 1.0$ Strong correlation
 $0.4 \leq |r| < 0.7$ Moderate correlation
 $|r| < 0.4$ Weak correlation
 $r = 0$ No linear correlation
 $r > 0$ Positive relationship
 $r < 0$ Negative relationship